

COMP 3311

DATABASE MANAGEMENT

SYSTEMS

LECTURE 4 EXERCISES

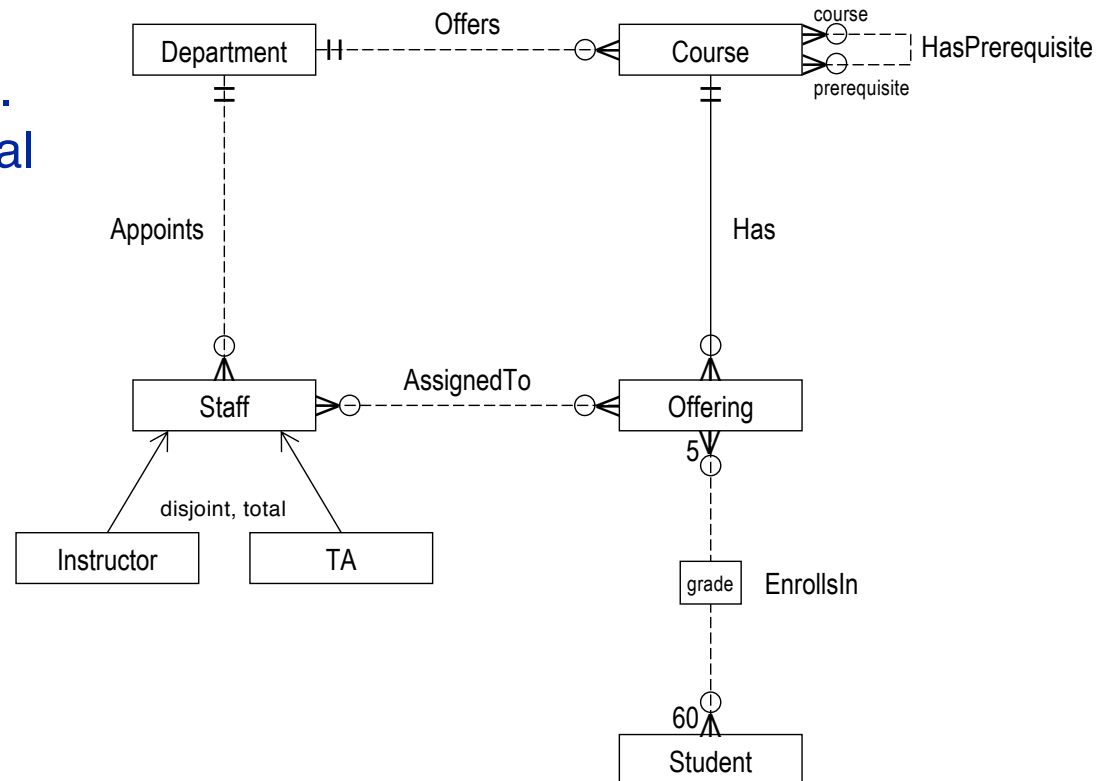
RELATIONAL DATABASE DESIGN:

E-R SCHEMA TO RELATION SCHEMAS REDUCTION

EXERCISE 1:

UNIVERSITY E-R SCHEMA REDUCTION

Reduce the university E-R schema to relation schemas. Specify all keys and referential integrity constraints.



Student
<u>studentId</u>
name
{major}

Department
<u>code</u>
name

Course
<u>courseId</u>
name

Offering
<u>section</u>
<u>semester</u>
<u>year</u>

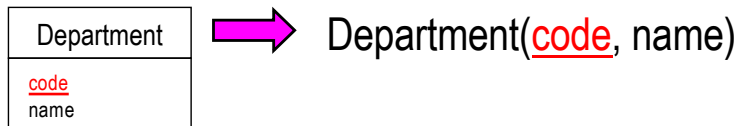
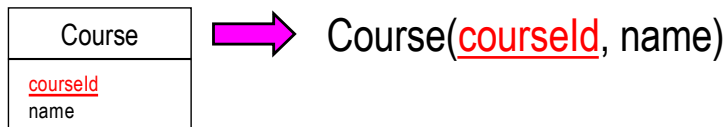
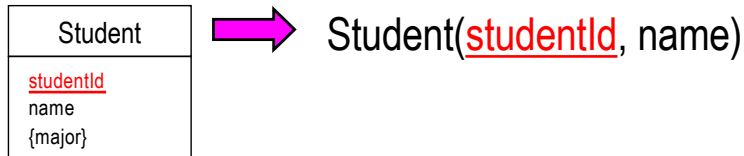
Staff
<u>hkid</u>
name
officeNumber

Instructor
title

TA

EXERCISE 1: REDUCE STRONG ENTITIES

Reduce **Student**, **Course** and **Department** strong entities



How do we reduce the strong entities?

⇒ Create a relation for each strong entity with the same attributes as the entity (excluding multivalued and composite attributes).

What are the keys of these relations? ⇒ Same as the entities.

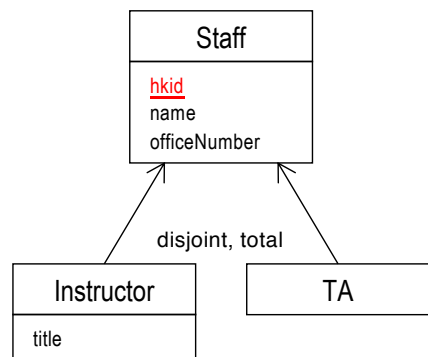
What are the foreign key constraints? ⇒ None.

EXERCISE 1: REDUCE GENERALIZATIONS

Reduce **Staff** generalization

Which option to select?

Option 1: Reduce all entities to relation schemas.

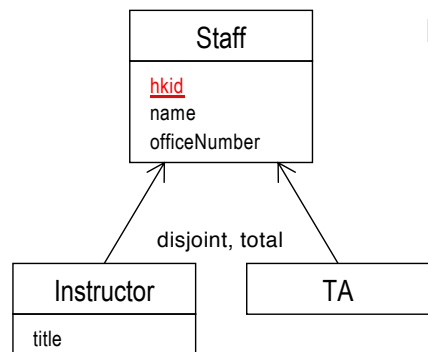


➡ Staff(hkid, name, officeNumber)

Instructor(hkid, title)
foreign key (hkid) references Staff(hkid)
on delete cascade

TA(hkid)
foreign key (hkid) references Staff(hkid)
on delete cascade

Option 2: Reduce only subtype entities to relation schemas.



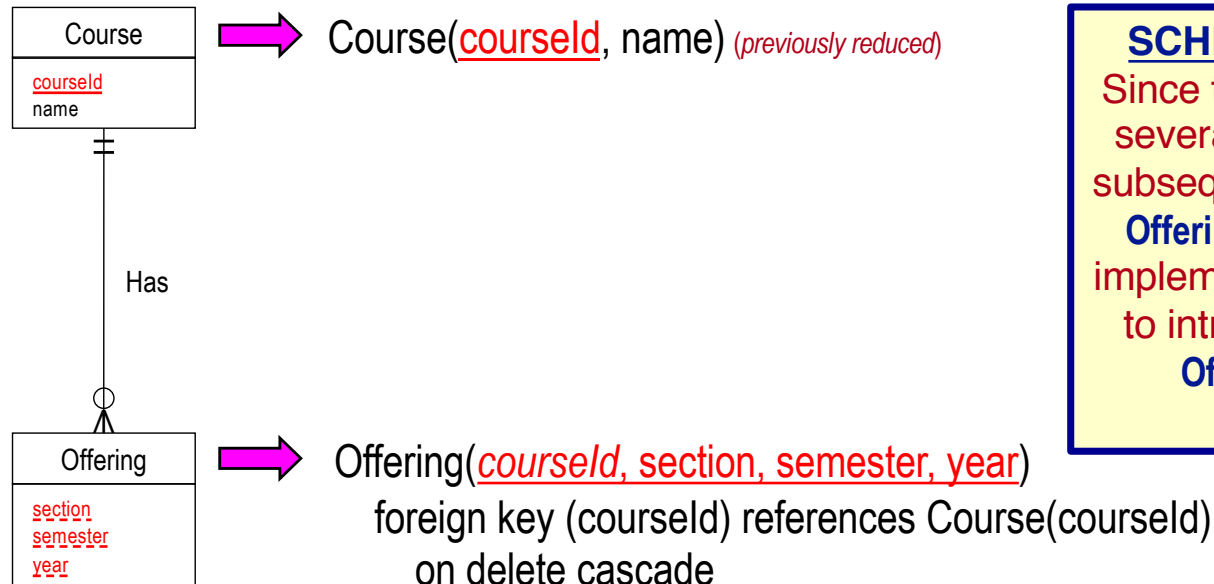
➡ Instructor(hkid, name, officeNumber, title)

TA(hkid, name, officeNumber)

Select Option 1 since only **Staff** has a relationship to other entities and some subtype entities have their own attributes.

EXERCISE 1: REDUCE WEAK ENTITIES

Reduce **Offering** weak entity together with **Has** identifying relationship



SCHEMA REDUCTION NOTE

Since the key for **Offering** contains several attributes, to simplify the subsequent reductions that involve **Offering** and to facilitate relation implementation, it would be helpful to introduce a surrogate key for **Offering** such as **offeringId**.
(Not done here.)

How do we reduce this entity?

⇒ Create a relation from **Offering** and include **courseId**, the key for **Course**, as a foreign key.

What is the key for this relation?

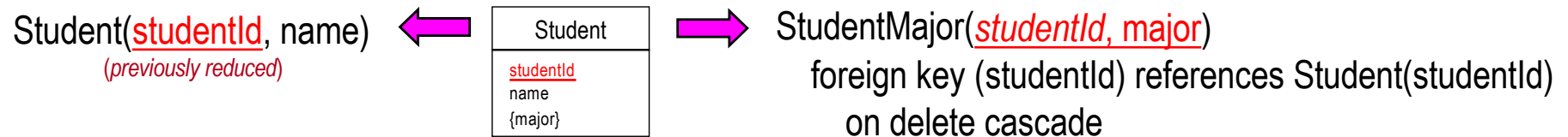
What is the foreign key constraint?

What is the referential integrity action?

EXERCISE 1:

REDUCE COMPOSITE/MULTIVALUED ATTRIBUTES

Reduce **major** multivalued attribute



How do we reduce the multivalued attribute **major**?

⇒ Create a relation **StudentMajor** and include **studentId**, the key for **Student**, and the attribute **major**.

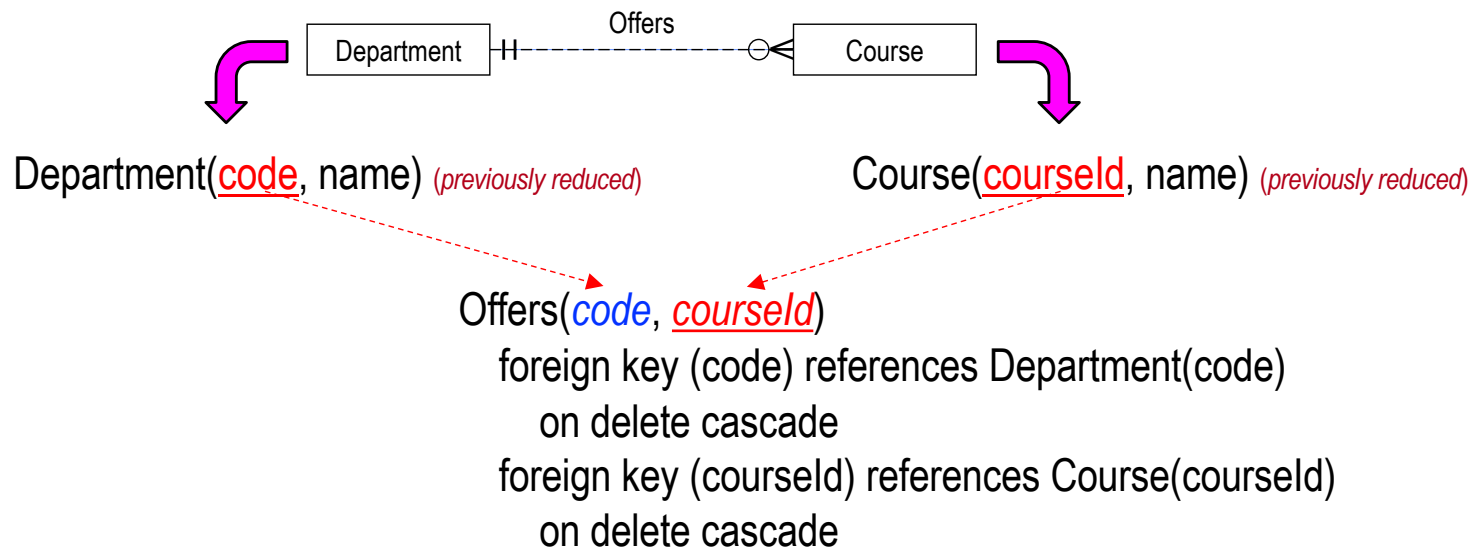
What is the key for this relation?

What is the foreign key constraint?

What is the referential integrity action—on delete cascade or on delete set null?

EXERCISE 1: REDUCE RELATIONSHIPS

Reduce **Offers** 1:N relationship between **Department** and **Course**



How do we reduce this relationship?

⇒ Create a relation, **Offers**, containing the keys for **Department** and **Course**.

What is the key for the relation?

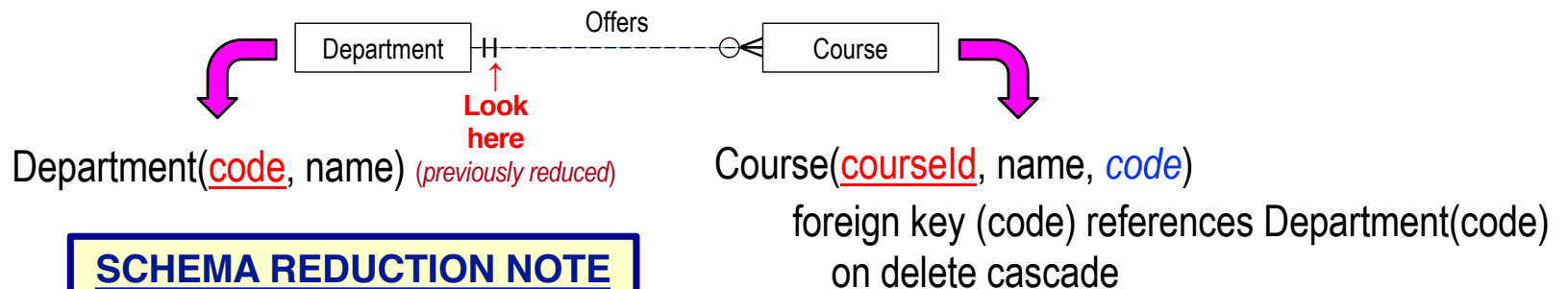
What are the foreign key constraints?

What are the referential integrity actions?

- **1:1 relationship** ⇒ the primary key for **either entity**.
- **1:N relationship** ⇒ the primary key for the **entity on the N-side** of the relationship.
- **N:M relationship** ⇒ the **union of the primary keys** for the two participating entities.

EXERCISE 1: REDUCE RELATIONSHIPS

Reduce **Offers** 1:N relationship between **Department** and **Course**
(using schema combination)



SCHEMA REDUCTION NOTE

The referential integrity action for the relation into which the foreign key is placed is determined by the participation constraint of the reduced entity.

- **partial**: on delete set null
- **total**: on delete cascade

Which relation do we use?

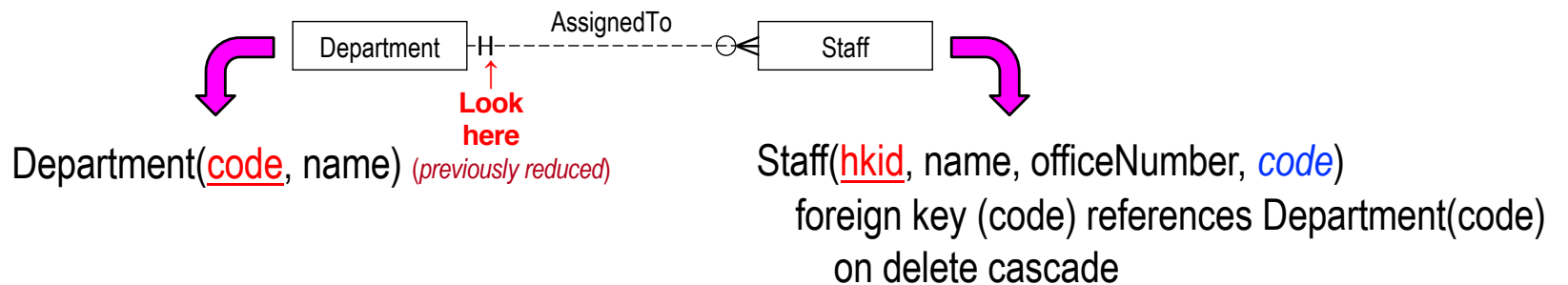
⇒ **Course** (Add **code**, the key for **Department**, as a foreign key.)

What is the foreign key constraint?

What is the referential integrity action?

EXERCISE 1: REDUCE RELATIONSHIPS

Reduce **Appoints** 1:N relationship between **Department** and **Staff**
(using schema combination)



Which relation do we use?

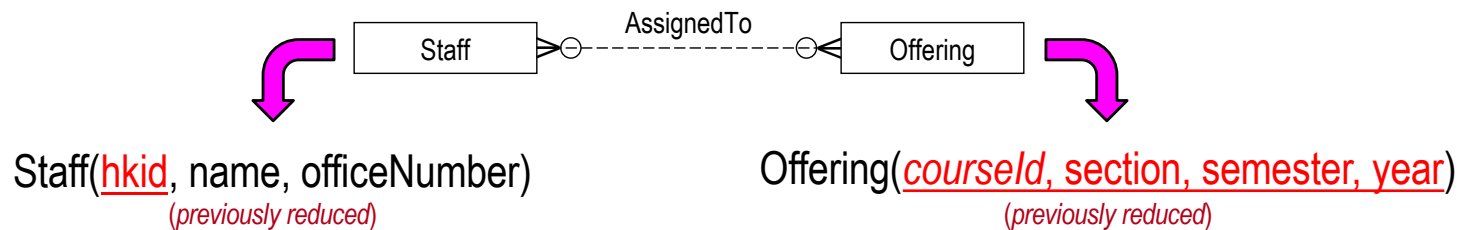
⇒ **Staff** (Add *code*, the key for **Department**, as a foreign key.)

What is the foreign key constraint?

What is the referential integrity action?

EXERCISE 1: REDUCE RELATIONSHIPS

Reduce **AssignedTo** N:M relationship between **Staff** and **Offering**



AssignedTo(hkid, courseId, section, semester, year)

foreign key (hkid) references Staff
on delete cascade

foreign key (courseId, section, semester, year) references Offering
on delete cascade

How do we reduce this relationship?

⇒ Create a relation, **AssignedTo**, containing the keys for the **Staff** and **Offering** relations.

What is the key for the relation?

What are the foreign key constraints?

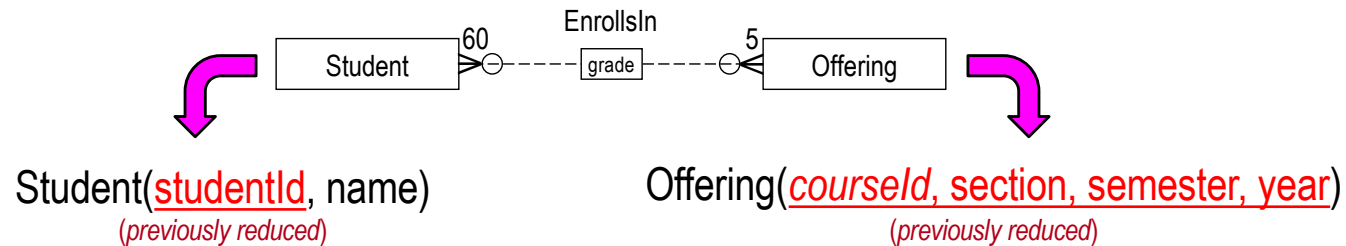
What are the referential integrity actions?

SCHEMA REDUCTION NOTE

For a relation that represents a relationship, the referential integrity action is always on delete cascade.

EXERCISE 1: REDUCE RELATIONSHIPS

Reduce **EnrollsIn** N:M relationship between **Student** and **Offering**



EnrollsIn(studentId, courseId, section, semester, year, grade)
foreign key (studentId) references Student(studentId)
on delete cascade
foreign key (courseId, section, semester, year) references
Offering(courseId, section, semester, year)
on delete cascade

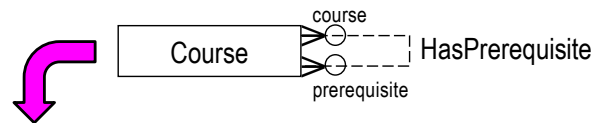
How do we reduce this relationship?

⇒ Create a relation, **EnrollsIn**, containing the keys for the **Student** and **Offering** relations.

Anything else? ⇒ Add the attribute **grade** to the relation.

EXERCISE 1: REDUCE RELATIONSHIPS

Reduce **HasPrerequisite** N:M relationship between **Course** and **Course**



Course(courseId, name) *(previously reduced)*

HasPrerequisite(courseId, prerequisiteId)

foreign key (courseId) references Course(courseId)
on delete cascade

foreign key (prerequisiteId) references Course(courseId)
on delete cascade

How do we reduce this relationship?

⇒ Create a relation, **HasPrerequisite**, containing the key for the **Course** relation (twice).

What is the key for the relation?

EXERCISE 1: UNIVERSITY RELATION SCHEMAS

Staff(hkid, name, officeNumber, code)

foreign key (code) references Department(code)
on delete cascade

Instructor(hkid, title)

foreign key (hkid) references Staff(hkid)
on delete cascade

TA(hkid)

foreign key (hkid) references Staff(hkid)
on delete cascade

Student(studentId, name)

Course(courseld, name, code)

foreign key (code) references Department(code)
on delete cascade

Department(code, name)

StudentMajor(studentId, major)

foreign key (studentId) references Student(studentId)
on delete cascade

Offering(courseld, section, semester, year)

foreign key (courseld) references Course(courseld)
on delete cascade

AssignedTo(hkid, courseld, section, semester, year)

foreign key (hkid) references Staff(hkid)
on delete cascade

foreign key (courseld, section, semester, year) references
Offering(courseld, section, semester, year)
on delete cascade

EnrollsIn(studentId, courseld, section, semester, year, grade)

foreign key (studentId) references Student(studentId)
on delete cascade

foreign key (courseld, section, semester, year) references
Offering(courseld, section, semester, year)
on delete cascade

HasPrerequisite(courseld, prerequisiteld)

foreign key (courseld) references Course(courseld)
on delete cascade

foreign key (prerequisiteld) references Course(courseld)
on delete cascade

REVIEW:

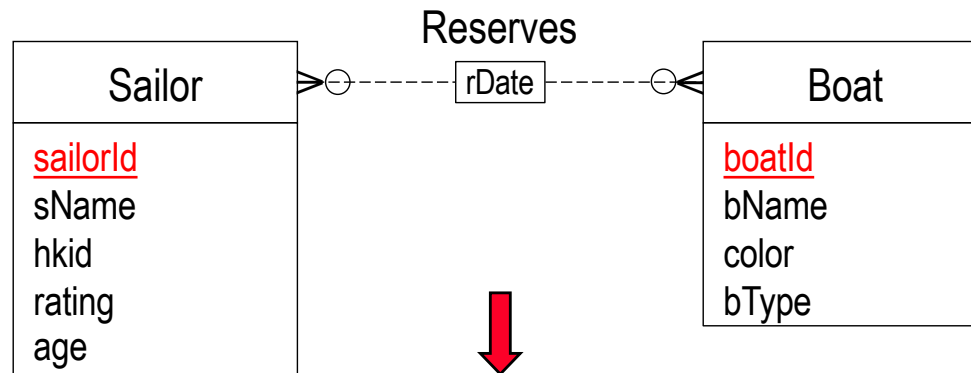
E-R SCHEMA TO RELATION SCHEMAS REDUCTION

Sailor(sailorId, sName, hkid, rating, age)

Boat(boatId, bName, color, bType)

Reserves(sailorId, boatId, rDate)

What is the E-R schema for these relation schemas?



What about this schema?

What do we get if we reduce the Reserves relationship?

Reserves(sailorId, boatId, rDate)

👉 **rDate is not part of the key in the reduction!**

What is the key?

👉 **A sailor can reserve a given boat at most once!**

REVIEW:

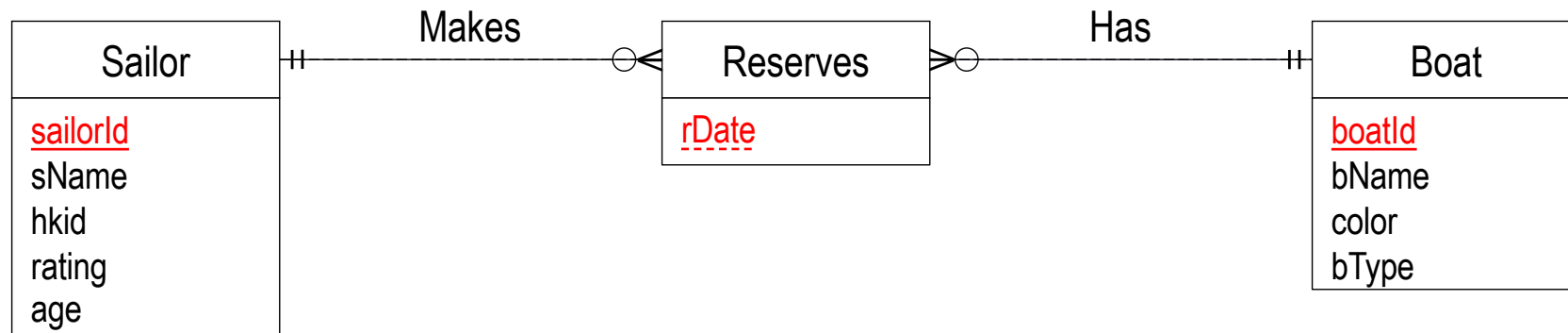
E-R SCHEMA TO RELATION SCHEMAS REDUCTION

Sailor(sailorId, sName, hkid, rating, age)

Boat(boatId, bName, color, bType)

Reserves(sailorId, boatId, rDate)

What about this schema?



What kind of entity is **Reserves**? $\Rightarrow$ weak entity

On which entity is **Reserves** dependent? $\Rightarrow$ both **Sailor** and **Boat**

Is rDate a discriminator for **Reserves**? $\Rightarrow$ probably—verify with client

What should be the cardinality constraints for **Makes**? $\Rightarrow$ 1:N

What should be the participation constraints for **Makes**? **Sailor** $\Rightarrow$ partial; **Reserves** $\Rightarrow$ total

What should be the cardinality constraints for **Has**? $\Rightarrow$ 1:N

What should be the participation constraints for **Has**? **Boat** $\Rightarrow$ partial; **Reserves** $\Rightarrow$ total

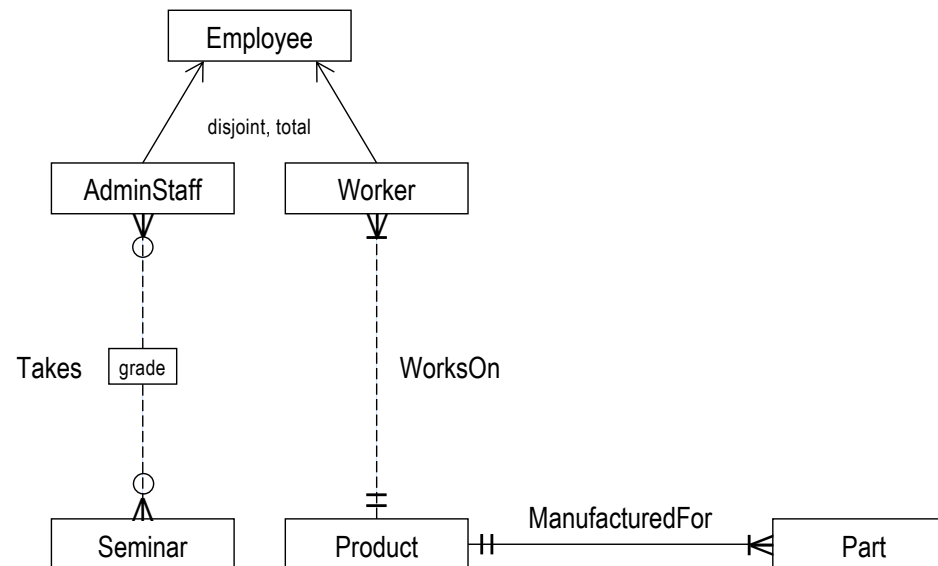
EXERCISE 2:

FACTORY APPLICATION E-R SCHEMA REDUCTION

Reduce the factory E-R schema to relation schemas.
Specify all referential integrity constraints and actions.

Do not add any surrogate keys.

Use schema combination where possible to reduce relationships.



Employee
<u>empld</u>
name
salary

AdminStaff

Worker

Seminar
<u>id</u>
name
date

Product
<u>id</u>
name

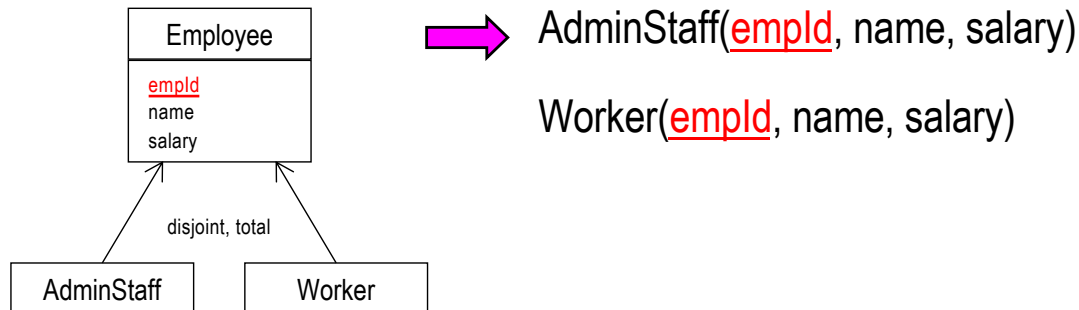
Part
<u>partNo</u>
color

EXERCISE 2:

REDUCE ENTITIES AND GENERALIZATIONS

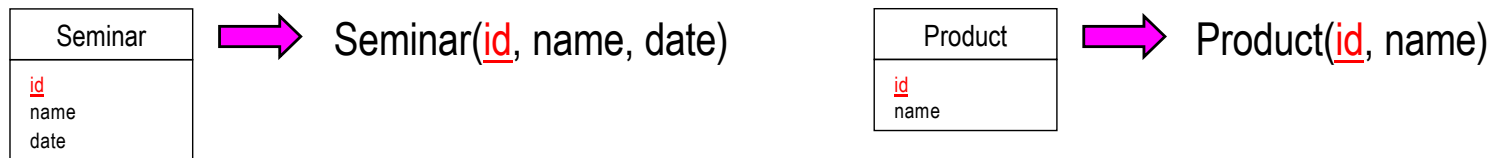
Reduce **Employee** generalization

Option 2: Reduce only subtype entities to relation schemas.

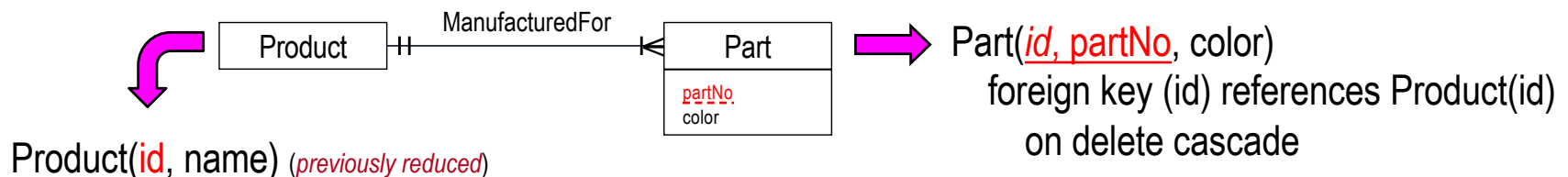


Select Option 2 since Employee has no relationships to other entities, the subtypes have no attributes, and the generalization is disjoint and total.

Reduce **Seminar** and **Product** strong entities

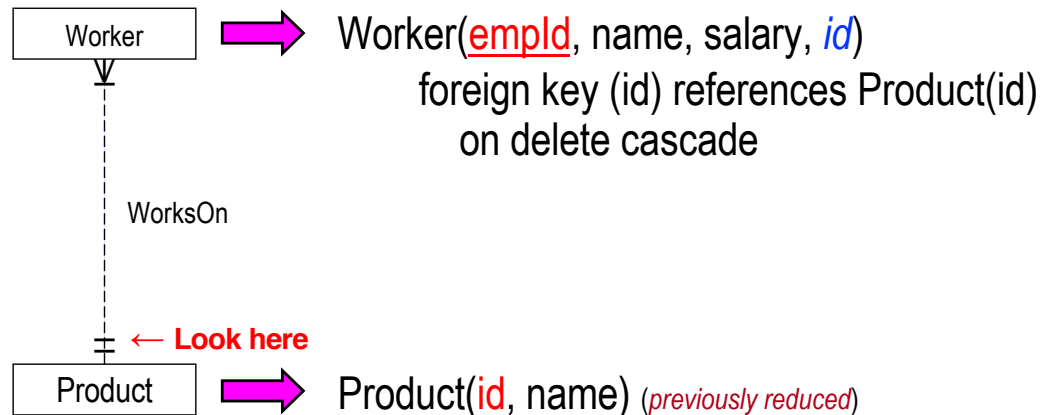


Reduce **Part** weak entity together with **ManufacturedFor** identifying relationship

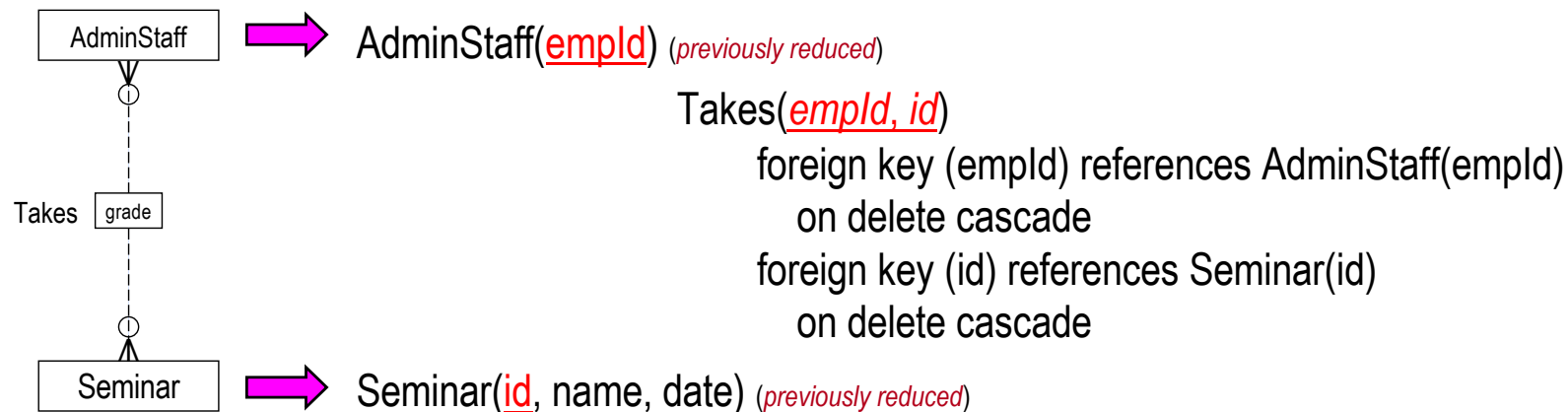


EXERCISE 2: REDUCE RELATIONSHIPS

Reduce **WorksOn** 1:N relationship between **Worker** and **Product** (using schema combination)



Reduce **Takes** N:M relationship between **AdminStaff** and **Seminar**



EXERCISE 2:

FACTORY APPLICATION RELATION SCHEMAS

AdminStaff(empld, name, salary)

Worker(empld, name, salary, id)
foreign key (id) references Product(id)
on delete cascade

Seminar(id, name, date)

Product(id, name)

Part(id, partNo, color)
foreign key (id) references Product(id)
on delete cascade

Takes(empld, id, grade)
foreign key (empld) references AdminStaff(empld)
on delete cascade
foreign key (id) references Seminar(id)
on delete cascade